

8.

Date ____/____/____

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Notebooks

- (1) There is an assumption that there is no significant difference between boys & girls with respect to intelligence.

using Two sample z-Test Tests are conducted on two groups and the following are the observations.

	Mean	S.D.	Size
Girls	89	4.82	50
Boys	82	9	120

Validate the claim with 5% L.O.S.

→ H_0 :- there is no diffⁿ between B & G

using the two-sample z-Test Formula

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - \mu_0}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}}$$

$$= \left(\frac{89 - 82}{\sqrt{\frac{16}{50} + \frac{81}{120}}} \right) = \left(\frac{7}{\sqrt{0.32 + 0.675}} \right)$$

$$= \frac{7}{\sqrt{0.995}} = \frac{7}{0.9974} = 7.018$$

At 5% level of significance

the critical z value is ± 1.96

since $7.018 > 1.96$ so we reject H_0

→

Date ___/___/___

② Analyze the below data and tell whether you can conclude that smoking causes cancer or not?

category	Diagnosed as cancer	without cancer	Total
Smokers	220	230	550
Non-smokers	350	640	990
Total	680	910	1590

→ H_0 = Smoking can't cause cancer
 H_1 = Smoking causes cancer

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

$$E_1 = \frac{550 \times 680}{1590} = 235.53 \rightarrow \frac{(O - E_1)^2}{E_1} = \frac{(220 - 235.53)^2}{235.53}$$

$$E_2 = \frac{550 \times 910}{1590} = 314.47 \rightarrow \frac{(O - E_2)^2}{E_2} = \frac{(230 - 314.47)^2}{314.47}$$

$$E_3 = \frac{990 \times 680}{1590} = 447.97 \rightarrow \frac{(O - E_3)^2}{E_3} = \frac{(350 - 447.97)^2}{447.97}$$

$$E_4 = \frac{990 \times 910}{1590} = 575.53 \rightarrow \frac{(O - E_4)^2}{E_4} = \frac{(640 - 575.53)^2}{575.53}$$

$$\chi^2 = 1.02 + 22.39 + 20.06 + 7.07$$

$$= 50.54$$

Degrees of freedom = $(C-1)(R-1)$

$$(2-1)(2-1) = 1$$

critical value at 5% level = 3.841

since $50.54 > 3.841 \rightarrow$ we reject the null hypothesis.